

BBE Sales Analytics

Data Cleaning & Transformation Documentation

Documentation of data preparation, validation and Power BI transformation steps

Item	Details
Source workbook	bbe_dataset (1).xlsx
Source tables	Products, Customers, Orders, Promotion
Final analytical tables	Dim_Product, Dim_Customer, Fact_Sales, Dim_Promotion, Dim_Date
Original Orders rows	58,178
Final Fact_Sales rows	58,154
Analysis period	January 2018 – January 2020

1. Purpose

This document records the data cleaning, validation and transformation work completed before building the BBE Sales Analytics Power BI dashboard. The objective was to create a reliable analytical dataset while preserving valid transactions and maintaining the original transaction grain.

2. Source Dataset

The source Excel workbook contains four business tables. Each table was prepared for its analytical role in the Power BI model.

Source table	Business purpose	Final Power BI name
Products	Product master and product attributes	Dim_Product
Customers	Customer master and customer attributes	Dim_Customer
Orders	Sales transaction lines	Fact_Sales
Promotion	Promotion definitions and discounts	Dim_Promotion

3. Transformation Naming & Structure

The Power Query queries were renamed to clearly communicate their role in the star schema. A dedicated date dimension was created for time intelligence and date-based reporting. Dim_Date was generated from the minimum and maximum OrderDate values and includes Year, Month, Quarter, Year Month and sorting attributes.

4. Products — Dim_Product

- Promoted headers and assigned appropriate numeric and text data types.
- Validated ProductKey for uniqueness and null values.
- Trimmed and cleaned relevant text fields.
- Preserved legitimate missing optional product attributes.
- Used ProductKey as the primary analytical key; ProductAlternateKey was not used as the primary key.
- Renamed the query from Products to Dim_Product.

5. Customers — Dim_Customer

- Promoted headers and assigned appropriate data types.
- Validated CustomerKey for uniqueness and null values.
- Cleaned relevant text fields.
- Removed Title because it was overwhelmingly null and added little analytical value.
- Preserved legitimate customer attribute nulls.
- Renamed the query from Customers to Dim_Customer.

6. Promotions — Dim_Promotion

- Assigned appropriate data types and cleaned promotion text.
- Validated PromotionKey for uniqueness and completeness.
- Retained unused promotions in the dimension.
- Renamed the query from Promotion to Dim_Promotion.

7. Orders — Fact_Sales

- Promoted headers and corrected transaction date, key and financial data types.
- Identified and removed exact duplicate transaction rows.
- Investigated and recovered five missing ProductKey values.
- Validated ProductKey, CustomerKey and PromotionKey against their dimensions.
- Validated chronological consistency of OrderDate, ShipDate and DueDate.
- Checked financial fields for invalid negative or zero values.
- Renamed the query from Orders to Fact_Sales.

8. Duplicate Transaction Handling

The original Orders table contained 58,178 rows. A duplicate check identified 24 exact duplicate rows. Because these records were exact duplicates rather than separate legitimate transaction lines, one copy of each was retained and the duplicate copies were removed.

Validation point	Result
Original Orders rows	58,178
Exact duplicate rows removed	24

Validation point	Result
Final Fact_Sales rows	58,154

9. Missing ProductKey Recovery

Five sales records initially contained missing ProductKey values. Instead of deleting those sales, the records were matched using transaction-level identifiers and sales information.

Sales Order	Line	Recovered ProductKey
SO75120	2	491
SO75120	3	225
SO75121	1	528
SO75121	2	537
SO75121	3	214

Result: Missing ProductKeys before cleaning = 5; after recovery = 0.

10. Relationship Validation

Relationship	Validation result
Fact_Sales[ProductKey] → Dim_Product[ProductKey]	Valid
Fact_Sales[CustomerKey] → Dim_Customer[CustomerKey]	Valid
Fact_Sales[PromotionKey] → Dim_Promotion[PromotionKey]	Valid

11. Date Validation

- OrderDate, ShipDate and DueDate were checked for chronological consistency.
- The sales period runs from 01-Jan-2018 to 27-Jan-2020.
- A dedicated Dim_Date table was created for Year, Month, Quarter and time-intelligence analysis.
- OrderDate is the active date relationship used for the main sales analysis.

12. Numeric Validation

Financial and numeric fields were reviewed for invalid negative or zero values. The primary sales fact fields reviewed were UnitPrice, ProductStandardCost, SalesAmount and TaxAmt. Product-level StandardCost, ListPrice and Weight were also checked.

13. Final Validation Checklist

Validation	Final status
Exact duplicate sales rows	0 remaining
Missing ProductKey	0 remaining
Invalid ProductKey	None identified
Invalid CustomerKey	None identified
Invalid PromotionKey	None identified
Invalid sales values	None identified
Final Fact_Sales records	58,154
Date range	Jan 2018 – Jan 2020

14. Final Analytical Structure

The cleaned data feeds a star-schema Power BI model consisting of Fact_Sales plus Product, Customer, Promotion and Date dimensions. This structure separates transaction data from descriptive attributes and supports consistent filtering, KPI calculations and dashboard analysis.

Final structure: Dim_Date → Fact_Sales ← Dim_Product, Dim_Customer and Dim_Promotion.

15. Conclusion

The cleaning process produced a validated 58,154-row sales fact table suitable for Power BI analysis. Duplicate transaction records were removed, missing product keys were recovered, relationships were validated, data types were standardized and unnecessary fields were removed. The resulting dataset provides a reliable foundation for the BBE Sales Analytics dashboard and accompanying business analysis.